

Name: _____

TOPIC TEST

The normal distribution

- Time allowed: 45 minutes
- Part A: 20 multiple-choice questions (20 marks)
- Part B: 9 free-response questions (30 marks)
- Total: 50 marks

Part A

20 multiple-choice questions

1 mark each: 20 marks

Circle the correct answer.

- 1 The lengths of nails at a factory are normally distributed with a length of 6 cm and a standard deviation of 0.05 cm. What percentage of nails have lengths less than 6.05 cm?

A 16% **B** 32%
C 68% **D** 84%

- 2 In a major city, the heights of adult females are normally distributed with a mean of 160 cm and a standard deviation of 8 cm. What percentage of females in this city would have a height between 152 cm and 176 cm?

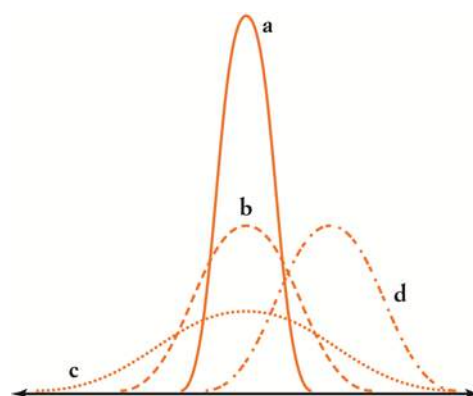
A 5% **B** 68%
C 81.5% **D** 99.7%

- 3 A machine is used to fill drink bottles at a factory. The volumes of drink in the bottles are found to be normally distributed with a mean of 1.5 L and a standard deviation of 0.05 L.

A bottle containing a volume that is more than 3 standard deviations away from the mean is rejected for sale. Which one of these volumes would be rejected?

A 1.32 L **B** 1.38 L
C 1.58 L **D** 1.62 L

- 4 Which statement is true about the distributions shown in this graph?



- A** The mean of **a** is greater than the mean of **b**.
B The mean of **d** is less than the mean of **a**.
C The mean of **c** is less than the mean of **b**.
D The mean of **a** is less than the mean of **d**.

- 5 Which statement is false about the distributions shown in the graph in question 4?

A The standard deviation of **a** is less than the standard deviation of **b**.
B The standard deviation of **d** is more than the standard deviation of **c**.
C The standard deviation of **c** is more than the standard deviation of **a**.
D The standard deviation of **b** is less than the standard deviation of **c**.

- 6 The head circumference of a population of infants is normally distributed with a mean of 49.5 cm and a standard deviation of 1.5 cm. In a sample of 800 boys taken from this population, how many would have a head circumference greater than 48 cm?

A 20 **B** 128
C 544 **D** 672

- 7 A herd of giraffes has a mean mass of 898 kg with a standard deviation of 83 kg. Within which interval would you expect 95% of giraffes to weigh?

A 732 kg to 981 kg
B 732 kg to 1064 kg
C 815 kg to 1064 kg
D 803 kg to 981 kg

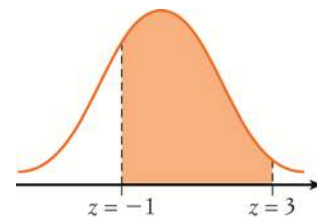
- 8 The annual rainfall in Hilltown is normally distributed with a mean of 1020 mm and a standard deviation of 260 mm. What is the z -score for a year in which 630 mm of rain fell?

A -2 **B** -1.5
C 1.5 **D** 2

- 9 The ages of bingo players at a club are normally distributed with a mean of 70 and a standard deviation of 6. If Monty's age has a z -score of -1.5, how old is he?

A 61 **B** 64
C 76 **D** 79

- 10 What percentage of scores lie in the shaded region under this normal curve?



A 15.85%
B 68%
C 83.85%
D 95%

- 11 The foot lengths of a population of Year 12 students are normally distributed with a mean of 24.5 cm. Jordan, a Year 12 student, has a foot length of 35 cm, which has a z -score of 2.5. What is the standard deviation for the foot lengths in the Year 12 population?

A 2.1 cm **B** 3 cm
C 4.2 cm **D** 10.5 cm

- 12** The number of online friends of a group of teenagers is shown on the stem-and-leaf plot below. What is the median of the data?

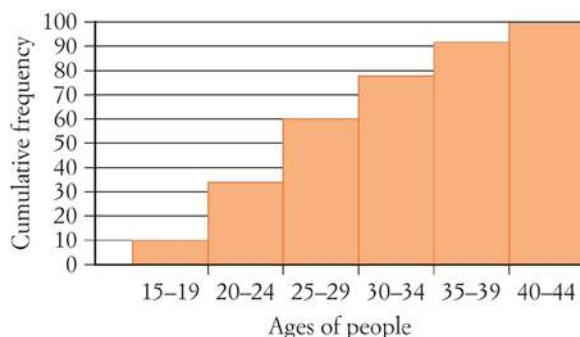
Steam	Leaf
1	0 4 4 6
2	1 3 6 8 9
3	2 3 5 7
7	9

- A** 26 **B** 27
C 28 **D** 24

- 13** What is the interquartile range of the data in question 12?

- A** 16 **B** 17
C 18 **D** 19

For questions 14 to 16, refer to this cumulative frequency histogram of the age groups of viewers of the TV show *Big Brother Theory*.



- 14** What was the median age of viewers?

- A** 20-24
B 25-29
C 30-34
D 35-39

- 15** What is the modal age of viewers?

- A** 20-24
B 25-29
C 30-34
D 40-44

- 16** What is the interquartile range?

- A** 8 **B** 9
C 10 **D** 12

- 17** Which statement is true about standard deviation?

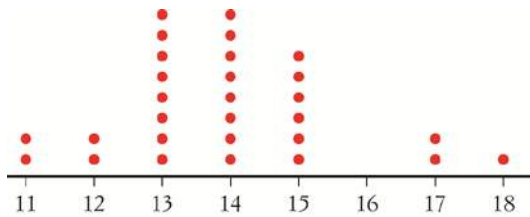
- A** It is a measure of central tendency that uses every score in its calculation.
B It is a measure of central tendency that uses some scores in its calculation.
C It is a measure of spread that uses every score in its calculation.
D It is a measure of spread that uses some scores in its calculation.

- 18** What is the standard deviation of this data set?

8 8 8 10 11 13 14 14 22

- A** 4.18 **B** 4.25
C 12 **D** 4.24

- 19** This dot plot shows the number of hours spent driving of a group of L-platers in a fortnightly period.



Which statement is true?

- A** The data is negatively skewed with a cluster around 13 to 15.
- B** The data is positively skewed with one mode.
- C** The data is positively skewed with a cluster around 13 to 15.
- D** The data is bimodal and negatively skewed.

- 20** The distances (in kilometres) that a small sample of students travel to get to school in a regional town are shown.

3, 3, 4, 5, 8, 10, 12, 15, 17, 32

What effect does ignoring the outlier have?

- A** The mean and median decrease and the standard deviation increases.
- B** The mean and median decrease, but the standard deviation remains the same.
- C** The mean, median and standard deviation all decrease.
- D** The mean and standard deviation decrease, but the median increases.

Part B

9 free-response questions

30 marks

Show all working

- 21** The masses of free-range eggs laid at a farm are normally distributed with a mean of 57 g and a standard deviation of 6 g. Between which 2 weights do 95% of the eggs lie?

[2 marks]

- 22** The skull widths of an adult population are normally distributed with a mean of 1395 mm.

a What is the standard deviation if 2.5% of skull widths are greater than 1493 mm?

b In a group of 600 adults, how many would have a skull width between 1346 mm and 1542 mm?

[4 marks]

- 23** The measured volumes of milk in 1000 mL (one litre) cartons are normally distributed with a mean of 1000 mL and a standard deviation of 8 mL.

a What is the z -score for a carton of milk containing:

i 1016 mL?

ii 988.8 mL?

b What volume of milk has a z -score of:

i 1.5?

ii -2.8?

c Cartons containing less than 992 mL are rejected for sale. What percentage of cartons are rejected?

[5 marks]

- 24** The scores on a Mathematics test are normally distributed with a mean of 62 and a standard deviation of 10. The scores on a Science test for the same class are normally distributed with a mean of 65 and a standard deviation of 8.

a Jack scored 74 on his Mathematics test. What is his z -score on this test?

b Jack also scored 74 on his Science test. What is his z -score on this test?

c On which test did Jack perform better?

d If 25 students sat for the Mathematics test, how many can be expected to score less than 72?

[5 marks]

- 25** Some results for Jean-Paul's French test and Sofia's Spanish test are shown below. Sofia and Jean-Paul performed equally well relative to their class. What did Sofia score in her Spanish test?

	Score	$\bar{x}$	Standard deviation
Jean-Paul	88	73	5
Sofia	?	68	4

[2 marks]

- 26** Keiji timed how long he took to solve each Sudoku puzzle he completed, correct to the nearest minute. The results are shown in the table below.

Time (min)	Frequency (f)	fx	Cumulative frequency
15	1		
16	5		
17	8		
18	8		
19	3		
20	2		
Total			

- a** How many games did Keiji play?

- b** Complete the table to calculate the mean, correct to one decimal place.

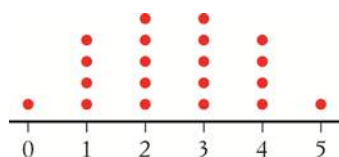
- c** What is the median?

- d** What is the interquartile range?

- e** Calculate the standard deviation, correct to one decimal place.

[5 marks]

- 27** A sample of Year 12 students was surveyed on how many different sports they played regularly. The results are shown on this dot plot.



- a** Describe the shape of the distribution.

- b** Find the median.

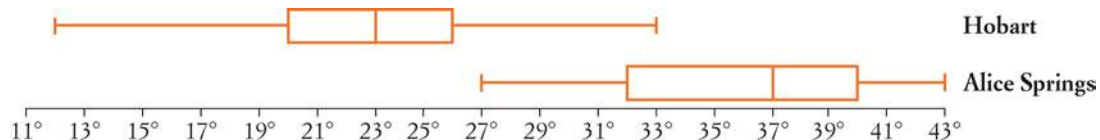
- c** Find the interquartile range.

[3 marks]

- 28** List 6 scores that have a mode of 6 and a median of 6.5.

[1 marks]

- 29** These parallel box plots describe the daily maximum temperatures in Hobart and Alice Springs for December.



- a** Which city has the higher interquartile range of temperatures? Show your working.

- b** Which city has a symmetrical distribution of temperatures?

- c** What was the difference in the median temperatures of Alice Springs and Hobart?

[3 marks]

This is the end of the test.

Use the back of this page for extra working space.

Answers

Part A

- | | | | | |
|-------------|-------------|-------------|-------------|-------------|
| 1 D | 2 C | 3 A | 4 D | 5 B |
| 6 D | 7 B | 8 B | 9 A | 10 C |
| 11 C | 12 B | 13 B | 14 B | 15 B |
| 16 C | 17 C | 18 D | 19 C | 20 C |

Part B

- 21** 45 g to 69 g
- 22** **a** 49 mm **b** 503
- 23** **a** **i** 2 **ii** -1.4
b **i** 1012 mL **ii** 977.6 mL
c 16%
- 24** **a** 1.2 **b** 1.125 **c** Mathematics **d** 21
- 25** 80
- 26** **a** 27 **b** 17.5 minutes **c** 17 minutes **d** 1 **e** 1.2
- 27** **a** symmetrical **b** 2.5 **c** $3.5 - 1.5 = 2$
- 28** Teacher to check, for example, 3, 6, 6, 7, 10, 14.
- 29** **a** Alice Springs **b** Hobart **c** 14°